A Hamilton-Jacobi approach for the evolutionary dynamics of populations in time-periodic environments

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joint work with Sepideh MIRRAHIMI

Winter School: CALCULUS OF VARIATIONS AND PROBABILITY
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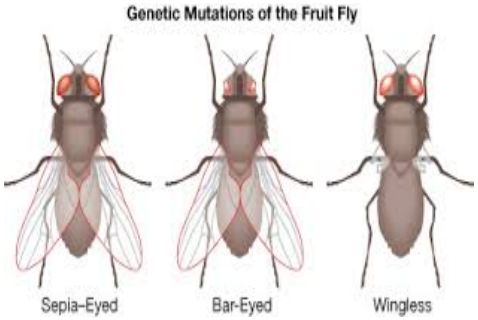
February, 2019

Motivations and Questions

We study the dynamics of a phenotypically structured population under the effect of mutations-selection in a fluctuating environment.



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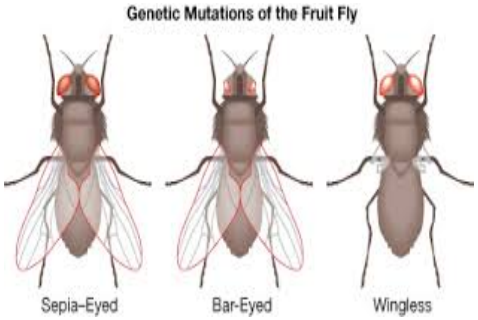

What will be the impact of the variations of the environment on the population's phenotypical distribution?
Periodic variations: seasonal effects, prescription of drugs, ...

Biological experience (Ketola et al. 2013)

Evolution of a population of bacterial pathogen *Serratia marcescens*. Several populations of bacteria kept in environments with constant or fluctuating temperature for several weeks and measured their growth rates in different environments.

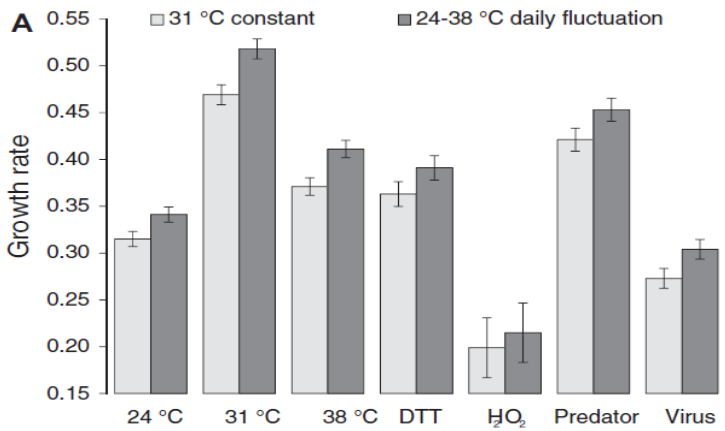

Figure: Average maximal growth rate of Serratia marcescens-bacteria evolved either in constant $31°$C or in daily fluctuating temperature between $24°$C and $38°$C, (Figure from Ketola et al. 2013)

Model and Notation

A parabolic integro-differential model with $(t,x) \in [0,+\infty) \times \mathbb{R}^d$ with non-local competition

$$\begin{cases} \partial_t n(t,x) - \varepsilon^2 \Delta n(t,x) = n(t,x)[a(e(t),x) - \rho(t)], \\ \rho(t) = \int_{\mathbb{R}^d} n(t,x)dx, \\ n(t=0,x) = n_0(x). \end{cases} \tag{P}$$

- x: phenotypic trait
- $n(t,x)$: density of trait x
- $\rho(t)$: total size
- $a(e(t),x)$: net growth rate
- $e(t)$: environmental state
- ε^2: effect of mutations

Assumptions

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- $0 \leq n_0(x) \leq e^{C_1 - C_2|x|}$

Convergence to a periodic solution

Proposition

For small values of ε *we have*

$$|\rho(t) - \rho_\varepsilon(t)| \to 0, \quad \|n(t,\cdot) - n_\varepsilon(t,\cdot)\|_{L^\infty} \to 0, \quad \textit{as } t \to \infty,$$

for $(n_\varepsilon, \rho_\varepsilon)$ *the unique periodic solution of*

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Expected asymptotic expansion: $u_\varepsilon(t,x) = u + \varepsilon v + O(\varepsilon^2)$.

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    (...) *widely extended to more complex models*

Summary

1. Introduction
2. Main result
3. Biological Examples
4. Heuristics and Idea of the proof

The Main Result

Theorem

(i) *As* $\varepsilon \to 0$*:*

$$n_\varepsilon(t,x) - \widetilde{\varrho}(t)\delta(x - x_m) \rightharpoonup 0,$$

where $\widetilde{\varrho}(t)$ *is the unique periodic solution of*

$$\frac{d\widetilde{\varrho}}{dt} = \widetilde{\varrho}[a(e(t), x_m) - \widetilde{\varrho}].$$

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$$\frac{d\widetilde{\varrho}}{dt} = \widetilde{\varrho}[a(e(t), x_m) - \widetilde{\varrho}].$$

(ii) *As* $\varepsilon \to 0$, $u_\varepsilon(t,x) = \varepsilon \ln n_\varepsilon(t,x)$ *converges locally uniformly to* $u(x)$*, the unique viscosity solution of:*

$$\begin{cases} -|\nabla u|^2 = \overline{a}(x) - \overline{a}(x_m), & x \in \mathbb{R}^d, \\ \max\limits_{x \in \mathbb{R}^d} u(x) = u(x_m) = 0. \end{cases}$$

(classical solution for $x \in \mathbb{R}$ *explicitly computed).*

Next order terms and approximations of the moments of the population's distribution

We are able to compute (formally) the next order term v which allows to obtain the following approximation of the population's distribution:

$$n_\varepsilon(t,x) \approx \frac{1}{(2\pi\varepsilon)^{d/2}} e^{\frac{u(x)}{\varepsilon}+v(t,x)}.$$

The Laplace's method for integration permit to obtain analytic formula for the moments of the population's distribution with an error of order ε^2 by mean of a Taylor expansion of u and v around x_m.

Notations

Mean Total population: $\overline{\rho} = \frac{1}{T}\int_0^T \rho_\varepsilon(t)dt$

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Variance: $\sigma^2(t) = \dfrac{1}{\rho_\varepsilon(t)}\displaystyle\int_{\mathbb{R}^d} (x-\mu)^2 n_\varepsilon(t,x)dx$

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Mean Fitness: $F(\tau) = \int_{\mathbb{R}^d} a(e(\tau),x)\frac{1}{T}\int_0^T \frac{n_\varepsilon(t,x)}{\rho_\varepsilon(t)}dtdx$

Oscillations on the optimal trait

- Periodic growth rate:

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$$\mu_p(t) \approx \frac{2\varepsilon c}{b}\sqrt{g}\sin\left(bt - \frac{\pi}{2}\right), \quad \sigma_p^2 \approx \frac{\varepsilon}{\sqrt{g}}, \quad \overline{\rho}_\varepsilon \approx r - \frac{gc^2}{2} - \varepsilon\sqrt{g},$$

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- Comparison of Fitness for population evolved in different environments:

$$F_p(\pi/b) \approx r - \varepsilon\sqrt{g} = F_c(\pi/b)$$

Oscillations on the pressure of the selection

- Periodic growth rate:

$$a(e(t), x) = r - g(e(t))x^2, \quad r > 0, \quad 0 \leq g(e(t)) : 1-\text{periodic}.$$

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$$F_p(\tau) \approx r - \varepsilon\frac{g(1/2)}{\sqrt{\overline{g}}} > F_c = r - \varepsilon\sqrt{g\,(1/2)}. \quad \text{for } \overline{g} > g(1/2)$$

Heuristics

Let $u_\varepsilon = \varepsilon \ln n_\varepsilon$, then for $(t,x) \in [0,+\infty) \times \mathbb{R}^d$:

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Formally asymptotic expansion for $u_\varepsilon, \rho_\varepsilon$ with $T-$periodic coefficients:

$$u_\varepsilon(t,x) = u(t,x) + \varepsilon v(t,x) + \varepsilon^2 w(t,x) + o(\varepsilon^2),$$

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give for u:

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and for v

$$\begin{cases} \partial_t v &= a(e(t),x) - \overline{a}(x) - \rho(t) + \overline{\rho}, \\ -\Delta u &= \dfrac{2\nabla u}{T}\displaystyle\int_0^T \nabla v dt - \overline{\kappa}. \end{cases}$$

Some relevant elements in the proof

1. Regularity estimates based on comparison principle and Bernstein type method.
   * Difficulty: Because of the periodicity we have not information on the initial conditions.
2. Convergence to the limit equation using a perturbed test function argument.
3. Uniqueness of viscosity solution of the Hamilton-Jacobi equation thanks to:
   * constraint of negativity on u
   * $\max a$ attained in a unique point

Conclusions and future work

What we have done

- In long time, the solution converges to the unique periodic solution of the problem.
- When mutations are small the solution concentrates on a single Dirac mass, while the size of the population varies periodically in time.
- We obtain good approximations for the distribution's moments of the population.

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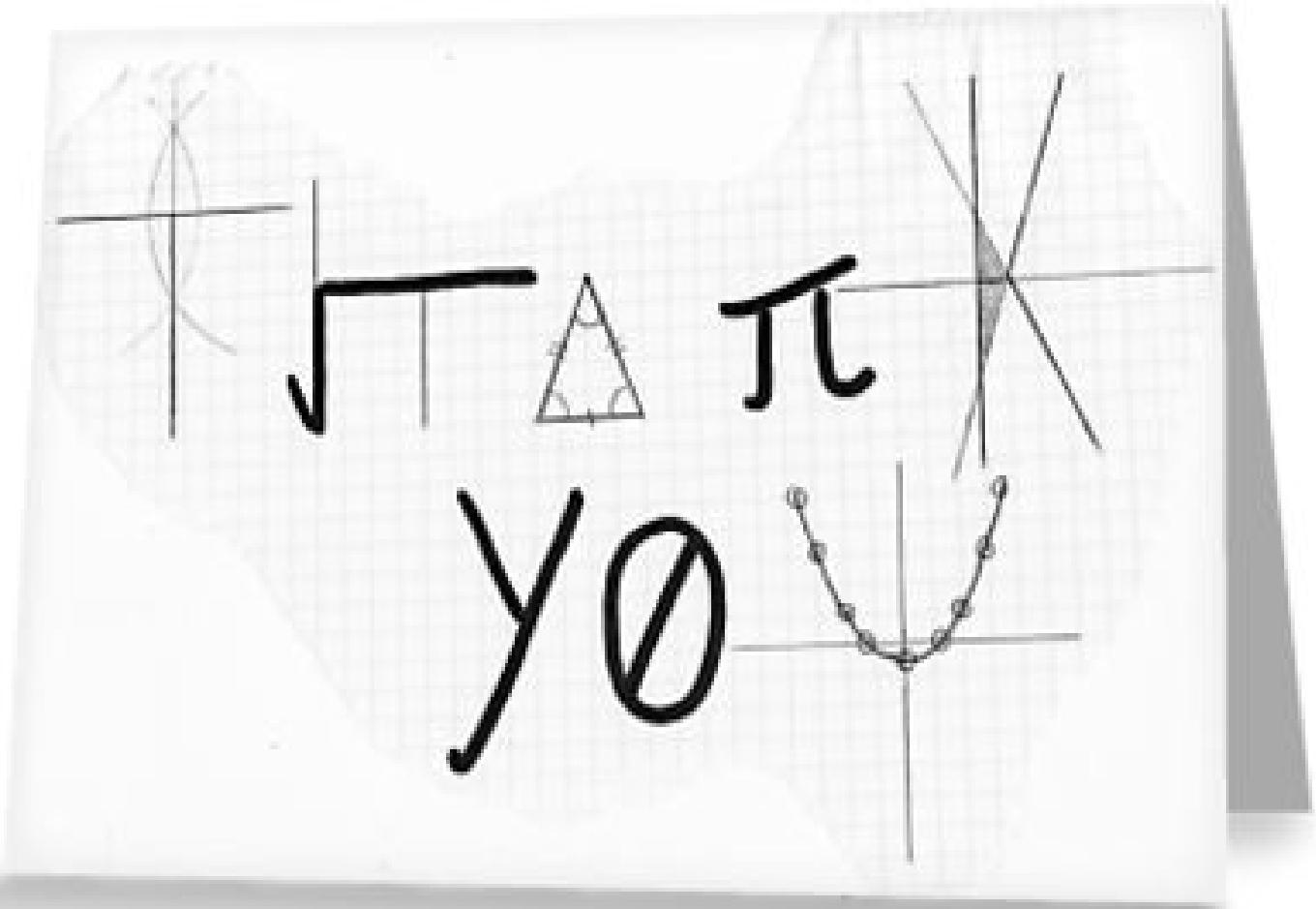
What we are doing and perspectives

- Study of the impact of a climate shift: $a(e(t), x - ct)$.
- Rigorous computation of next order terms: $v(t, x)$.
- Study of the solution when a has several maxima.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